MALAPPURAM SAHODAYA APTITUDE TEST
EXAMINATION 2023-24
DISTRICT LEVEL

Class: 11(PCM) **Date: 4/11/23** **Marks: 100**

Name of the Student: **Time: 2 Hrs**

General Instructions:

1. *Candidate will be supplied with separate Question Paper and Answer Sheet.*
2. *Answer the questions in the OMR sheet by shading the appropriate bubble.*
3. *Shade the bubbles with black ball pen/ HB pencil.*
4. *Question number **1 to 100** carries one mark each.*
5. *For each question below, four options are given. One of them is correct answer. Make yourchoice and shade the correct box.*

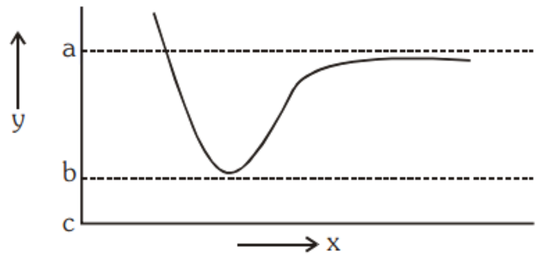
1. In a college of 300 students, every student reads 5 newspaper and every newspaper is read by 60 students. The number of newspapers is
   a. At least 30 b. At most 20 c. Exactly 25 d. Exactly 60
2. If X and Y are two sets and X' denotes the compliment of X, then $X \cap (X \cup Y)'$ is equal to
   a. X b. Y c. ϕ d. $X \cap Y$
3. Domain of $\sqrt{(a^2 - x^2)}$ $(a > 0)$ is
   a. $(-a, a)$ b. $[-a, a]$ c. $[0, a]$ d. $(-a, 0]$
4. The value of $\tan 3A - \tan 2A - \tan A$ is equal to
   a. $\tan 3A \tan 2A \tan A$
   b. $-\tan 3A \tan 2A \tan A$
   c. $\tan A \tan 2A - \tan 2A \tan 3A - \tan 3A \tan A$
   d. None of these
5. If $\tan A = \frac{a}{b}$ then $a \sin 2A + b \cos 2A$
   a. a
   b. b
   c. $\frac{a}{b}$
   d. None of these
6. If $x^n - 1$ is divisible by $x - k$ then the least positive integral value of k is
   a. 0 b. 1 c. 2 d. 3
7. The complex number z which satisfies the condition $\left|\frac{i+z}{i-z}\right| = 1$ lies on
   a. Circle $x^2 + y^2 = 1$
   b. The $x-$ $axis$
   c. The $y-$ $axis$
   d. The line $x + y = 1$
8. If $a + ib = c + id$, then
   a. $a^2 + c^2 = 0$
   b. $b^2 + c^2 = 0$
   c. $b^2 + d^2 = 0$
   d. $a^2 + b^2 = c^2 + d^2$
9. Given that x, y and b are real numbers and $x < y, b < 0$, then
   a. $\frac{x}{b} < \frac{y}{b}$ b. $\frac{x}{b} \leq \frac{y}{b}$ c. $\frac{x}{b} > \frac{y}{b}$ d. $\frac{x}{b} \geq \frac{y}{b}$
10. If $|x + 2| \leq 5$, then
   a. $x \in (-7,3)$
   b. $x \in [-7,3]$
   c. $x \in (-\infty, -7) \cup (3, \infty)$
   d. $x \in (-\infty, -7] \cup [3, \infty)$
11. The value of ${}^{2n}C_3 : {}^{n}C_3 = 12 : 1$?
   a. 5
   b. 6
   c. 4
   d. 10

12. Everybody in a room shakes hands with everybody else. The total number of handshakes is 190. The total number of persons in the room is
  a. 20 b. 11 c. 13 d. 60
13. The coefficient of x^n in the expansion of $(1+x)^{2n}$ and $(1+x)^{2n-1}$ are in the ratio?
  a. 1:2 b. 1:3 c. 3:1 d. 2:1
14. The coefficient of x^6y^3 in the expansion of $(x+2y)^9$ is
  a. $\frac{9!}{3!6!}$ b. $\frac{9!}{3!6!}2^6$ c. $\frac{9!}{3!3!}2^3$ d. $\frac{9!}{3!6!}2^3$
15. The ratio of the sum of m and n terms of an AP is $m^2{:}n^2$. The ratio of m^{th} and n^{th} terms is
  a. $m{:}n$
  b. $2m-1{:}2n-1$
  c. $m^2{:}n^2$
  d. $2m+1{:}2n+1$
16. The sum of the sequence 7,77,777, … … upto n terms is
  a. $\frac{7}{81}(10^{n+1}-9n-10)$
  b. $\frac{7}{9}(10^{n+1}-9n-10)$
  c. $\frac{7}{81}(10^{n}-9n-10)$
  d. $\frac{7}{81}(10^{n+1}+9n-10)$
17. If p and q are lengths of perpendicular from the origin to lines $xcos\theta - ysin\theta = kcos2\theta$ and $xsec\theta + ycosec\theta = k$, then
  a. $p^2+q^2=k^2$
  b. $p^2+2q^2=k^2$
  c. $p^2+4q^2=k^2$
  d. $p^2+4q^2=2k^2$
18. A line passes through (1,2) such that its intercept between the axes is bisected at P. The equation of the line is
  a. $x+2y=5$
  b. $x-y+1=0$
  c. $x+y-3=0$
  d. $2x+y-4=0$
19. The angle between the lines $\sqrt{3}x+y=1$ and $x+\sqrt{3}y=1$ is
  a. $\frac{\pi}{6}$ b. $\frac{\pi}{4}$ c. $\frac{\pi}{2}$ d. $\frac{\pi}{3}$
20. The sum of first p terms of an AP is equal to the sum of the first q terms, then the sum of first $(p+q)$ terms is
  a. $p+q$ b. $p-q$ c. $-p+q$ d. 0
21. The right option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40)
  [$CaCO_3 \rightarrow CaO + CO_2$]
  a. 1.76 g b. 2.64 g c. 1.32 g d. 1.12
22. The maximum number of atoms is present in which of the following -
  a. 1 g of Mg b. 1 g of C c. 1 g of Li d. 1 g of Ag
23. An organic compound contains 80% (by wt.) carbon and the remaining percentage of hydrogen. The right option for the empirical formula of this compound is
  a. CH_3 b. CH_4 c. CH d. CH_2
24. What will be the pressure of the gaseous mixture when 0.5 L of H_2 at 0.8 bar and 2.0 L of dioxygen at 0.7 bar are introduced in a 1L vessel at 27°C?
  a. 0.4 bar b. 1.4 bar c. 1.8 bar d. 2.8 bar
25. Given below are the critical temperatures of a few gases. When the gases are started cooling, which gas will liquefy first and which will liquefy in the end?

Gas	N2	CO2	NH3	O2
TC/K	126	304	405	154

  a. N_2 will liquefy first and NH_3 at last
  b. NH_3 will liquefy first and CO_2 at last
  c. NH_3 will liquefy first and N_2 at last
  d. CO_2 will liquefy first and NH_3 at last
26. Which of the following is not a correct postulate of kinetic theory of gases?
a. The molecules of a gas are continuously moving in different directions with different velocities.
b. The average kinetic energy of the gas molecules is directly proportional to the absolute temperature of the gas.
c. The volume of the gas is due to the large number of molecules present in it.
d. The pressure of the gas is due to the collision of the molecules on the walls of the container.

27. A nitrogen laser produces radiation at a wavelength of 337.1 nm. If the number of photons emitted is 5.6×10^{24}, the power of the laser is:
    a. 4.56×10^{7} J
    b. 3.33×10^{6} J
    c. 1.29×10^{6} J
    d. 9.17×10^{5} J
28. The energy of an electron in the first Bohr's orbit of an H-atom is -13.6 eV. The possible energy value (s) of the excited state(s) for electrons in Bohr's orbits of hydrogen is (are):
    a. -3.4 eV
    b. -4.2 eV
    c. -6.8 eV
    d. +6.8 eV
29. The number of sigma (σ) and π bonds in pent -2−en−4−yne are
    a. 10 σ bonds and 3 π bonds
    b. 8 σ bonds and 5 π bonds
    c. 11 σ bonds and 2 π bonds
    d. 13 σ bonds and no π bonds
30. The potential energy (Y) curve for H_2 formation as a function of inter nuclear distance (X) of the H atoms is shown below.



The bond energy of H_2 is
    a. (c-a)
    b. (b-a)
    c. (b-a) / 2
    d. (c-a) / 2
31. BF_3 is planar and electron deficient compound. Hybridization and number of electrons around the central atom respectively are:
    a. Sp^2 and 6
    b. Sp^2 and 8
    c. Sp^3 and 4
    d. Sp^3 and 6
32. In both water and dimethyl ether (CH_3–O–CH_3) oxygen atom is the central atom and has the same hybridization, yet they have different bond angles. Which one has greater bond angle?
    a. Water < dimethyl ether
    b. Dimethyl ether > Water
    c. Dimethyl ether = Water
    d. Can not be determined.
33. Which of the following is used to oxidize ethanol to ethanoic acid?
    a. Alkaline $KMnO_4$
    b. Conc. H_2SO_4
    c. Acidified $K_2Cr_2O_7$
    d. All of above
34. Soaps are formed by saponification of
    a. Alcohols
    b. Glycosides
    c. Simple esters
    d. Carboxylic acids
35. The isomer pair of Hexyne is
    a. Cyclohexane
    b. Cyclohexene
    c. Benzene
    d. 2-Methylhexene
36. A projectile is thrown at a speed V and at an angle θ with the horizontal. If the speed at its maximum height is $\frac{V}{3}$, then the value of tan θ is:
    a. $\sqrt{3}$
    b. $\frac{1}{\sqrt{3}}$
    c. $2\sqrt{2}$
    d. 3
37. Consider a vector addition $\vec{P} + \vec{Q} = \vec{R}$. If $\vec{P} = |\vec{P}|\hat{\imath}$, $|\vec{Q}| = 10$ and $\vec{R} = 3|\vec{P}|\hat{\jmath}$, then $|\vec{P}|$ is:
    a. $\sqrt{10}$
    b. 30
    c. $\sqrt{30}$
    d. $2\sqrt{10}$
38. A car is moving with an initial speed of 5 m/s. A constant braking force is applied and the car is brought to rest in a distance of 10 m. What is the average speed of the car during the deceleration process?
    a. 1 m/s
    b. 2.5 m/s
    c. 4 m/s
    d. 5 m/s
39. Suppose a force is given by the expression = kx^2, where x has the dimension of length. The dimension of k is:
    a. $ML^{-1}T^{-1}$
    b. MLT^{-1}
    c. MT^{-2}
    d. $ML^{-1}T^{-2}$
40. A horizontal force is exerted on a 20 kg box to slide it up on an inclined plane with an angle of 30°. The frictional force retarding the motion is 80 N. If the box moves with a constant speed, then the magnitude of the force is: [Take g = $10 ms^{-2}$]

a. $50\sqrt{2}$ N b. 100 N c. $80\sqrt{3}$ N d. $120\sqrt{3}$ N

41. A uniform thin rod of mass 3 kg has a length of 1 m. If a point mass of 1 kg is attached to it at a distance of 40 cm from its centre, the centre of mass shifts by a distance of:
   a. 2.5 cm b. 5 cm c. 8 cm d. 10 cm
42. A planet has an escape speed of 10 km/s, The radius of the planet is 10,000 km. The acceleration due to gravity of the planet at its surface is:
   a. 10 m/s^2 b. 9.8 m/s^2 c. 20 m/s^2 d. 5 m/s^2
43. An ideal diatomic gas is made up of molecules that do not vibrate. Its volume compressed by a factor of 32, without any exchange of heat. If the initial and final pressures are P_1 and P_2, respectively, the ratio $P_1:P_2$, is:
   a. 1:128 b. 128:1 c. 1:32 d. 32:1
44. A glass capillary of radius 0.15 mm is dipped into a liquid of density and surface tension 1600 kg/m^3 and 0.12 Nm^{-1}, respectively. The liquid in the capillary rises by a height of 5.0 cm. The contact angle between liquid and glass will be: [Take g = 10 ms^{-2}]
   a. 30° b. 60° c. 45° d. 75°
45. A billiard ball B_1, moving with velocity V, collides with another billiard ball After the collision, ball B_2, is deflected by 60° and the angle between the velocities of these two balls is 90°. The speed of the ball B_2 after the collision is:
   a. $\frac{V}{2}$ c. $\frac{\sqrt{3}V}{2}$
   b. $\frac{3V}{2}$ d. $\frac{2V}{\sqrt{3}}$
46. The kinetic energy of a particle of mass m_1 moving with a speed V is same as the kinetic energy of a solid sphere of mass m_2, rolling on the plane surface. If the speed of the centre of the sphere is also V, then $\frac{m_1}{m_2}$ is:
   a. $\frac{7}{10}$ c. $\frac{5}{7}$
   b. $\frac{1}{2}$ d. $\frac{7}{5}$
47. A spherical ball is subjected to a pressure of 100 atmosphere. If the bulk modulus of the ball is 10^{11} N/m^2, then change in the volume is:
   a. 10^{-1} % b. 10^{-2} % c. 10^{-3} % d. 10^{-4} %
48. A heat engine operates between a cold reservoir and a hot reservoir. The engine takes 200 J of heat from the hot reservoir and has the efficiency of 0.4. The amount of heat delivered to the cold reservoir in a cycle is:
   a. 100 J b. 120 J c. 140 J d. 160 J
49. A simple pendulum experiment is performed for the value of 'g', the acceleration due to the earth's gravity. The measured value of length of the pendulum is 25 cm with an accuracy of 1 mm and the measured time for 100 oscillations is found to be 100 sec with an accuracy of 1 sec. The percentage in the determination of 'g' is:
   a. 9.8 b. 0.98 c. 4.8 d. 2.4
50. A metal wire of natural length 50 cm and cross-sectional area 4.0 mm^2 is fixed at end. A mass of 2.4 kg is hung from the other end of the wire. If the elastic potential energy of the wire is 1.8 x 10^{-4} J, then its Young's modulus is: (Take g = 10 ms^{-2})
   a. 1.6 x 10^{11} Nm^{-2} b. 2.4 x 10^{11} Nm^{-2} c. 3.2 x 10^{11} Nm^{-2} d. 2.0 x 10^{11} Nm^{-2}
51. Choose the one which can be substituted for the given words/phrase: Not likely to be easily pleased
   a) Fastidious b) Infallible c) Fatalist d) Communist
52. Find the proper order for the four sentences and mark accordingly:
   1: The lions used to be widely found in Africa and Asia.
   P: There are special forest zones set aside for wildlife in various countries.
   Q: Indiscriminate killing by hunters has been the cause of this drastic fall in their number.
   R: Today they are a relatively rare species.
   S: If the species survives at all, it will be only in national parks.
   6: No hunting is permitted in such reserved areas.
   a) Q S P R b) R S P Q c) S R P Q d) R Q S P

53. Punctuate the following sentence: great god I would rather be a pagan than such a christian wrote wordsworth.
a) 'Great God! I would rather be a Pagan than such a Christian,' wrote Wordsworth
b) “Great God! I would rather be a Pagan than such a Christian,” wrote Wordsworth.
c) “Great God! I would rather be a Pagan than such a Christian”, wrote Wordsworth.
d) Great God! I would rather be a Pagan than such a Christian, wrote Wordsworth

54. Choose the most suitable option to fill in the blank: Son: I failed, father. I don't understand how. Father condemned: __________
a) Well, never mind. Try again next year.
b) Well, I do! You are lazy.
c) Perhaps there has been a mistake.
d) You really care that you should pass.

55. Fill in the blank with the correct option: The earliest known Greek _______ are Mycenaean, written on clay tablets, and ________ records, but cannot be _________ literature
a) runics, document, termed
b) writings, document, termed
c) document, register, termed
d) transcripts, classify, termed

56. Read the passage and answer the question: In front of the house, there was a beach. Peter liked to stretch on the sand when the sun was warming it. From the back door of the house, he would walk along a path as far as the sand, and stand at the water's edge looking at the sea. When nothing very interesting was happening on the water, he would go down on his knees and take a handful of sand. Through his fingers ran the stand until only small stones and shells were left. Then, with a large sweep of his arm and with as much strength as he could muster, Peter would throw the shells away as far as he could. At other times, he would go on his back and gaze up at the clouds, his hands idly searching the sand at his sides. He never stopped playing with the sand and feeling it run through his fingers, however much he was absorbed by the changes in the clouds. Occasionally, some fishing boats came close enough to the beach for Peter to see what the fishermen were doing. Then with his hands clasped, he would look and look, while his whole body moved with the boat from side to side. When the fisherman drew in their nets or cast them into the water, Peter would do the same with an imaginary net of his own his place on the beach. On this small beach, Peter had a world of his own. When nothing interesting was happening on the water, Peter would______________.
a) take a handful of sand and let it run through his fingers
b) watch the fishing boats
c) pretend to be a fisherman by playing with his imaginary net
d) go back to the house

57. Choose the word which best expresses the meaning of the word given in the quotes: We were “ecstatic” but increasingly nervous.
a) Animated
b) Bewildered
c) Enraptured
d) Wilful

58. In the following question, a sentence has been given in Active/Passive Voice. Out of the four alternatives suggested, select the one which best expresses the same sentence in Passive/Active Voice. "Are you upset about your exam result?" asked a soft voice from behind.
a) A soft voice asked that if I was upset about my exam result.
b) A soft voice behind me asked if I was upset about my exam result.
c) A soft voice said to me are you upset about my exam result.
d) A soft voice from my back asked if I was upset about my exam result.

59. Read the following passage and answer the question: Manik Pradhan, a power-loom owner from Maharashtra, and his mother were homeward-bound one evening when heavy rain forced them to take shelter beneath a bridge. Not far away, a small group of laborers huddled together under a part of the cement housing above a 16-meter-deep well used to pump water for irrigation. Suddenly, Manik and his mother heard the laborers scream. When the two got to the well, they were told that a five-year-old boy named Hariya had fallen in through a side opening in the structure. Ignoring his mother's fears, Manik quickly knotted together lengths of flimsy rope belonging to the laborers and asked them to lower him into the dark well. "I hope the rope holds," he thought. As he descended, Manik noticed the metal rungs on the wall of the well. He grabbed hold of one

and started climbing down when he saw the boy clinging to a pipe running up the well's center. Grabbing the child, Manik started to climb praying that the old rungs wouldn't give away and plunge them both into the churning water below. Their luck held and within a few minutes, Manik clambered to ground level and handed over Hariya to his sobbing father. The man fell at Manik's feet and offered him some money as a reward. Refusing the cash, Manik took Hariya and his family to a nearby eatery and offered them steaming tea to warm them up. Several organizations have honored Manik for his bravery and presence of mind on that wet day three years ago. "I am happy I was at the right place at the right time," he says, "and was able to return a little boy to his family." On reading the above passage, from the given set of options choose the right option which first drew Manik and his mother to the well?

a) They wanted to take shelter from the heavy rain
b) They were returning back home from their journey
c) They heard news of about a small boy falling into the well
d) They heard screams of the labourers

60. Read the given statement and identify the best word that captures the expression of the father: She's scared to ask for pocket money as her father is in a black mood today.
a) Occupied b) Irritated c) Busy d) Worried

Direction (Q. No. 61 and 62) : Choose the correct option.

61. Raman _______ for reaching home late.
(A) took him to task (B) die in harness
(C) carried of his feet (D) faced the music

62. The light bulb ______. Please change it.
(A) broke off (B) burnt out (C) dealt with (D) carried on

63. 3. Choose one word that would fit in both the blanks.
(i) During the cyclone the _____ of the tree fell off.
(ii) The new bank has opened three ____ in rural areas.
(A) leaves (B) offices (C) branches (D) routes

64. Find two words with the same sound but different spelling for the given sentences.
I cannot ____ to see any animal suffering. The giant pulled the roof off the house with his ____ hands.
(A) bare,beer (B) bare, bear (C) bear, bare (D) bear, beer

Direction (Q. No. 65 and 66) : Read the extract and answer the following questions.

Department stores in the capital region, suffering from a prolonged slump in sales, are changing tactics by marketing new products and making their displays more interesting. Some stores are reducing the amount of floor space devoted to name brands in order to display more of their own house brands. Others are using existing floor space for their own speciality shops. Still others are targeting special segments of the population. Stores that are developing their own brands are able to offer quality clothing at 30 percent off name brand prices.

65. What has caused department stores to alter marketing strategies?
(A) Poor sales (B) New tax regulations (C) Lack of storage space (D) Production costs

66. Which strategy is not being implemented?
(A) Marketing to specific groups (B) Developing in-store brands
(C) Purchasing additional floor space (D) Making displays more interesting

67. Choose the correct meaning of the word in italics. The precocious lad quickly mastered German, Latin and principal Slavonic languages, frequently acting as his father's interpreter.
(A) Someone who is highly intelligent (B) A boy who cannot focus
(C) A fit boy (D) Someone who easily learns languages

68. Choose the incorrect part of the sentence.
(A) The government (B) has been working (C) on this project (D) from the last 20 years.

69. Choose the most appropriate option. Everyone should have periodic eye examinations to make sure problems are quickly _______.
(A) prepared (B) discover (C) responded (D) discovered

70. Reorder the jumbled sentences to make a cohesive paragraph.
a. On the other hand, my mother is not so tall; she is a little fat, has brown hair and a fair complexion. b. My

father is dark, very tall and of medium weight.
c. However, they look nice together. Our parents' differences start with their appearance.
d. They don't look similar at all.
(A) cdab (B) bacd (C) bdac (D) dbac

71. Fill in the blank with the present perfect tense form of the verb given in the bracket. "James ___________ to visit her grandmother." (go)
(A) have been going (B) had gone (C) has gone (D) has been going

72. Fill in the blank with the future perfect continuous tense form of the verb given in the bracket. "We __________ for an hour now." (wait)
(A) shall have been waiting (B) have been waiting
(C)shall be waiting (D) had been waiting

73. Choose the present indefinite tense form of the sentence. "He had been sleeping."
A. He has been sleeping.
B. He had slept.
C. He has slept.
D. He sleeps.

74. Identify the tense used in the underlined phrase. "He joined the cooking school late but only in a few months had mastered the recipes."
A. Past indefinite tense
B. Past perfect tense
C. Present perfect tense
D. Present indefinite tense

75. By the time she was fifteen, she __________ a beautiful singer.
A. shall become
B. become
C. had become
D. has becoming

76. Fill in the blank with the present perfect continuous tense form of the verb given in the bracket. "Adam __________ a health regime everyday." (follow)
A. follows B. have been following C. has been following D. has followed

77. Neither she nor I __________ at home.
A. were B. was C. has been D. be

78. My friends and I __________ stuck there.
A. had been B. was C. has been D. have had been

79. Choose the correct sentence.
A. Sam have gone to buy some groceries.
B. Sam had going to buy some groceries.
C. Sam be going to buy some groceries.
D. Sam is going to buy some groceries.

80. Which tense is used to express an action taking place in the future?
A. Future perfect tense
B. Future perfect continuous tense
C. Future indefinite tense
D. Future continuous tense

81. Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?
a. 13/90 b. 12/90 c. 78/90 d. 77/90

82. A deck of 5 cards (each carrying a distinct number from 1 to 5) is shuffled thoroughly. Two cards are then removed one at time from the deck. What is the probability that the two cards are selected with the number on the first card being one higher than the number on the second card?
a. 1/5 b. 4/25 c.1/4 d. 2/5

83. Choose the most appropriate word from the options given below to complete the following sentence: If we manage to ____________ our natural resources, we would leave a better planet for our children.
a. Uphold b. Restrain c. Cherish d. Conserve

84. Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean -1 and variance unknown If P(X <=-1) = P(Y >=2). the standard deviation of Y is
a. 3 b. 2 c. sqrt (2) d. 1

85. Choose the most appropriate word from the options given below to complete the following sentence. He could not understand the judges awarding her the first prize, because he thought that her performance was quite __________.
    a. Superb   b. Medium   c. Mediocre   d. Exhilarating
86. There is bag of lots of integers from 1 to 10000, but you can only pick the numbers that are of 3-digit and also divisible by 3. What maximum sum of numbers you can pick.
    a. 165150   b. 1600843   c. 168132420   d. 165322501
87. Sum of the age of 4 children born at interval of 4 years is 36. What is the age of youngest child?
    a. 2 years   b. 3 years   c. 4 years   d. 5 years
88. A is 5 years older than B who is thrice as old as C. If the total of ages of A, B and C is 40, then how old is C ?
    a. 6   b. 7   c. 5   d. 8
89. How many terms are there in 3,9,27,81........531441?
    a. 25   b. 12   c. 13   d. 14
90. If two fair coins are flipped and at least one of the outcomes is known to be a head, what is the probability that both outcomes are heads?
    a. 1/3   b. 1/4   c. 1/2   d. 2/3
91. A man travelled a certain distance by train at the rate of 25 kmph. and walked back at the rate of 4 kmph. If the whole journey took 5 hours 48 minutes, the distance was ?
    a. 25 km   b. 30 km   c. 20 km   d. 15km
92. A man is walking at a speed of 10 kmph. After every km, he takes a rest for 5 minutes. How much time will he take to cover a distance of 5 km ?
    a. 30 minutes   b. 40 minutes   c. 50 minutes   d. 60 minutes
93. The ages of X and Y are in the proportion of 6:5 and total of their ages is 44 years. The proportion of their ages after 8 years will be ?
    a. 6:3   b. 3:6   c. 9:5   d. 8:7
94. A train, 150 m long, takes 30 seconds to cross a bridge 500 m long. How much time will the train take to cross a platform 370 m long ?
    a. 36 secs   b. 30 secs   c. 24 secs   d. 18 secs
95. The mean of 50 observations was 36. It was found later that an observation 48 was wrongly taken as 23. The corrected new mean is ?
    a. 35   b. 36.5   c. 40   d. 42
96. The angle of elevation of a ladder leaning against a wall is 60 deg and the foot of the ladder is 4.6 m away from the wall. The length of the ladder is ?
    a. 2.3m   b. 4.6m   c. 9.2m   d. 7.8m
97. If Raj was one-third as old as Rahim 5 years back and Raj is 17 years old now, How old is Rahim now ?
    a. 36   b. 48   c. 40   d. 41
98. A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of his son is ?
    a. 2   b. 20   c. 21   d. 22
99. If DRIVE is coded as 59372, and SPUR is coded as 6489, what will be code for PRIDE ?
    a. 49235   b. 94532   c. 49352   d. 43925
100. In a code language, MACHINE is written as CAMHENI. How will MONSTER be written as in that language ?
    a. NOMSRET   b. OMNSETR   c. NOMETSR   d. SNOMRET

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